$$
\begin{aligned}
C_{\phi W}^{(21)} \to \frac{1}{16\,\pi^2}
\Bigg(& -\frac{1}{2}\, m_2\, \tilde{\mu}\, g_2{}^4\, LF_{2,2,0}[m_2,\, \tilde{\mu}] + m_2\, \tilde{\mu}\, g_2{}^4\, LF_{3,2,-1}[m_2,\, \tilde{\mu}] + m_2\, \tilde{\mu}\, g_2{}^4\, LF_{4,1,-1}[m_2,\, \tilde{\mu}] - \frac{1}{4}\, \tilde{\mu}\, g_2{}^2\, y_e{}^{pr}\, \overline{a_e}^{pr} \\
& LF_{4,1,-1}\left[m_{\tilde{l}}{}^p,\, m_{\tilde{e}}{}^r\right] + \frac{1}{4}\, \tilde{\mu}\, g_2{}^2\, y_e{}^{pr}\, \overline{a_e}^{pr}\, LF_{5,1,-2}\left[m_{\tilde{l}}{}^p,\, m_{\tilde{e}}{}^r\right] - \frac{3}{4}\, \tilde{\mu}\, g_2{}^2\, y_d{}^{pr}\, \overline{a_d}^{pr}\, LF_{4,1,-1}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{d}}{}^r\right] + \\
& \frac{3}{4}\, \tilde{\mu}\, g_2{}^2\, y_d{}^{pr}\, \overline{a_d}^{pr}\, LF_{5,1,-2}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{d}}{}^r\right] - \frac{3}{8}\, \tilde{\mu}\, g_2{}^2\, y_u{}^{pr}\, \overline{a_u}^{pr}\, LF_{2,2,0}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{u}}{}^r\right] - \\
& \frac{3}{8}\, \tilde{\mu}\, g_2{}^2\, y_u{}^{pr}\, \overline{a_u}^{pr}\, LF_{3,1,0}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{u}}{}^r\right] + \frac{3}{8}\, \tilde{\mu}\, g_2{}^2\, y_u{}^{pr}\, \overline{a_u}^{pr}\, LF_{3,2,-1}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{u}}{}^r\right] + \\
& \frac{3}{8}\, \tilde{\mu}\, g_2{}^2\, y_u{}^{pr}\, \overline{a_u}^{pr}\, LF_{4,1,-1}\left[m_{\tilde{q}}{}^p,\, m_{\tilde{u}}{}^r\right] + \frac{3}{8}\, \tilde{\mu}\, g_2{}^2\, y_u{}^{pr}\, \overline{a_u}^{pr}\, LF_{3,1,0}\left[m_{\tilde{u}}{}^r,\, m_{\tilde{q}}{}^p\right] + \\
& \frac{3}{8}\, \tilde{\mu}\, g_2{}^2\, y_u{}^{pr}\, \overline{a_u}^{pr}\, LF_{3,2,-1}\left[m_{\tilde{u}}{}^r,\, m_{\tilde{q}}{}^p\right] - \frac{9}{8}\, \tilde{\mu}\, g_2{}^2\, y_u{}^{pr}\, \overline{a_u}^{pr}\, LF_{4,1,-1}\left[m_{\tilde{u}}{}^r,\, m_{\tilde{q}}{}^p\right] + \\
& \frac{3}{4}\, \tilde{\mu}\, g_2{}^2\, y_u{}^{pr}\, \overline{a_u}^{pr}\, LF_{5,1,-2}\left[m_{\tilde{u}}{}^r,\, m_{\tilde{q}}{}^p\right] + \frac{1}{8}\, m_1\, \tilde{\mu}\, g_1{}^2\, g_2{}^2\, LF_{3,1,0}[\tilde{\mu},\, m_1] - \\
& \frac{1}{4}\, m_1\, \tilde{\mu}\, g_1{}^2\, g_2{}^2\, LF_{4,1,-1}[\tilde{\mu},\, m_1] + \frac{1}{4}\, m_1\, \tilde{\mu}\, g_1{}^2\, g_2{}^2\, LF_{5,1,-2}[\tilde{\mu},\, m_1] + \frac{3}{8}\, m_2\, \tilde{\mu}\, g_2{}^4\, LF_{3,1,0}[\tilde{\mu},\, m_2] + \\
& m_2\, \tilde{\mu}\, g_2{}^4\, LF_{3,2,-1}[\tilde{\mu},\, m_2] - \frac{3}{4}\, m_2\, \tilde{\mu}\, g_2{}^4\, LF_{4,1,-1}[\tilde{\mu},\, m_2] + \frac{3}{4}\, m_2\, \tilde{\mu}\, g_2{}^4\, LF_{5,1,-2}[\tilde{\mu},\, m_2]\Bigg)
\end{aligned}
$$